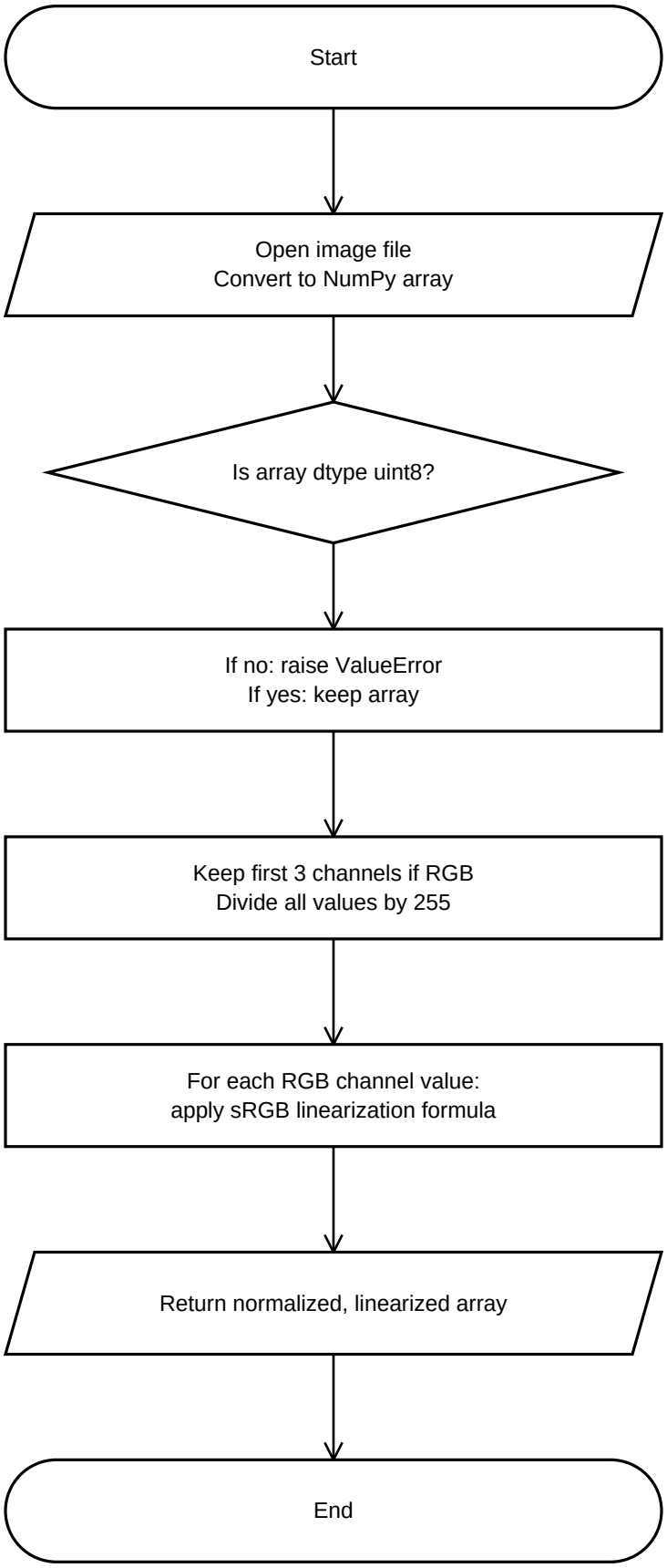
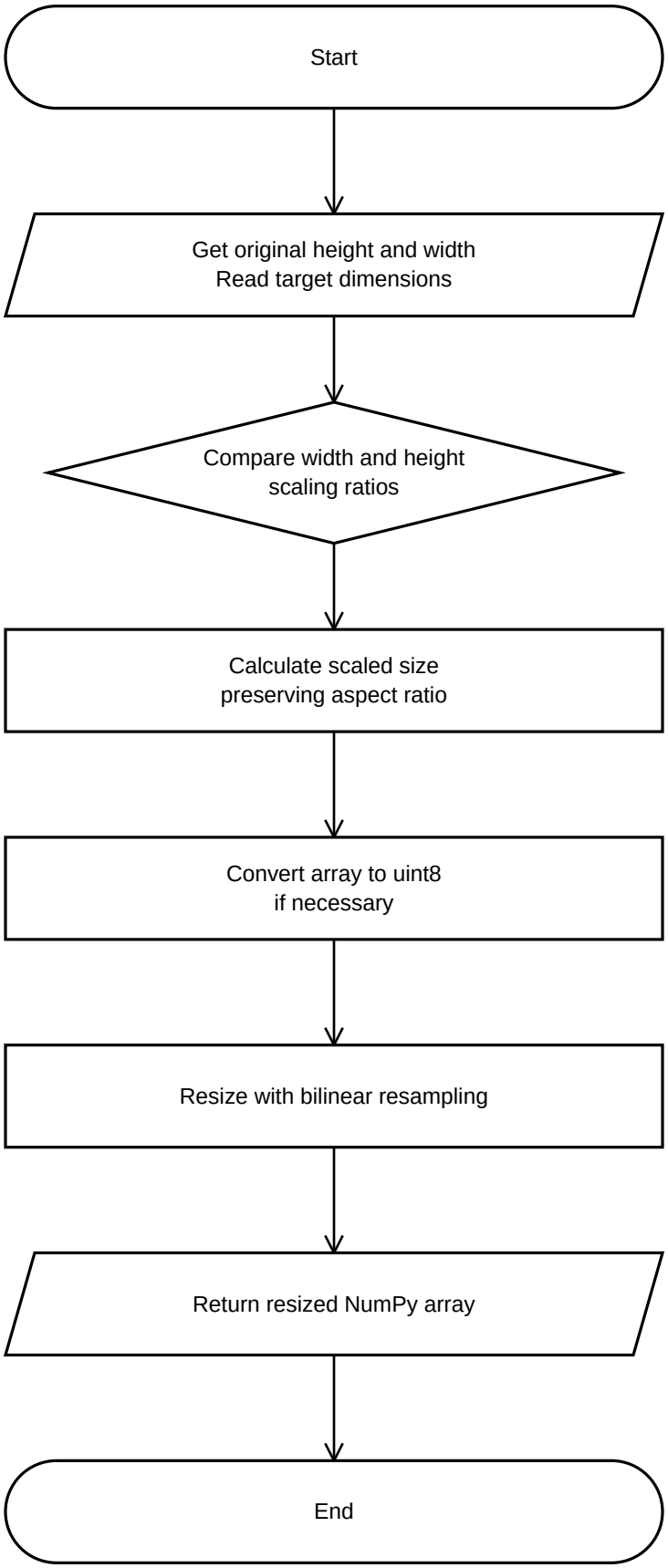
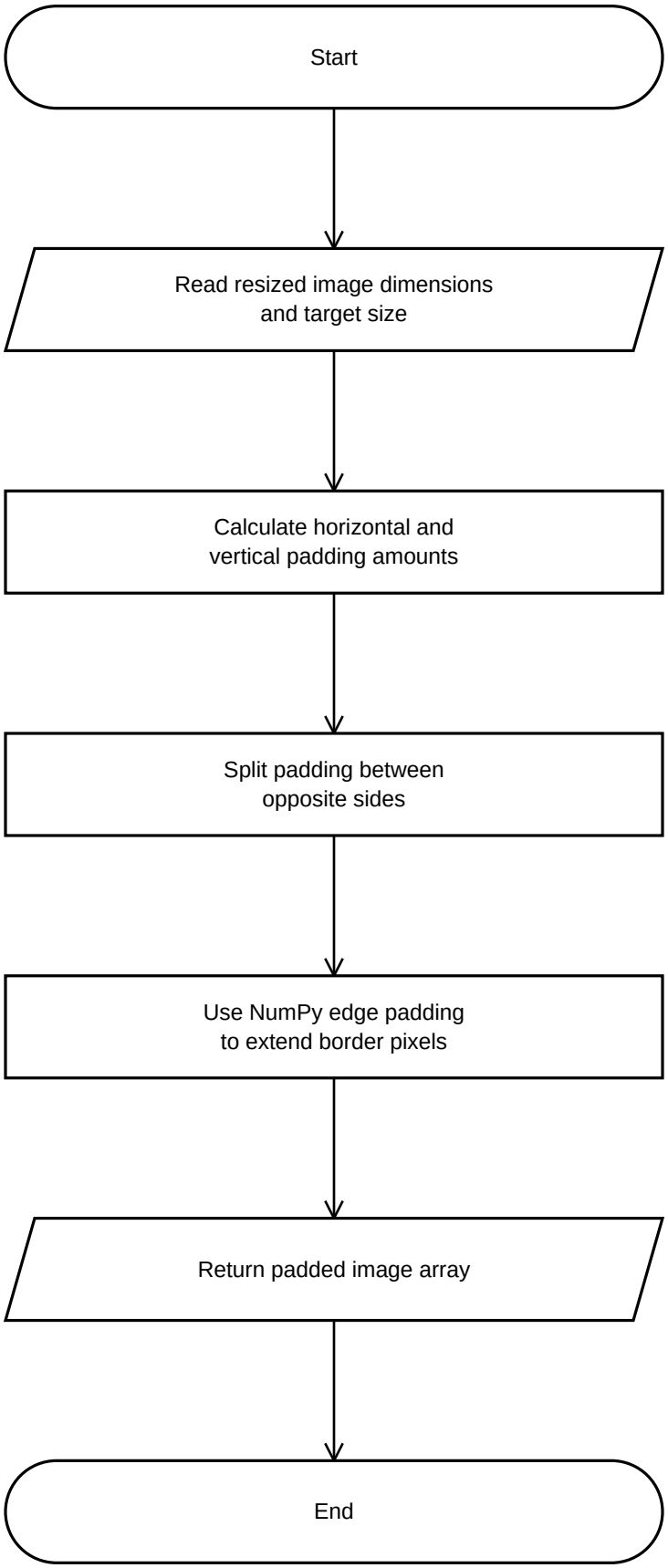
1. load_image


Start
Open image file
Convert to NumPy array
Is array dtype uint8?
If no: raise ValueError
If yes: keep array
Keep first 3 channels if RGB
Divide all values by 255
For each RGB channel value:
apply sRGB linearization formula
Return normalized, linearized array
End

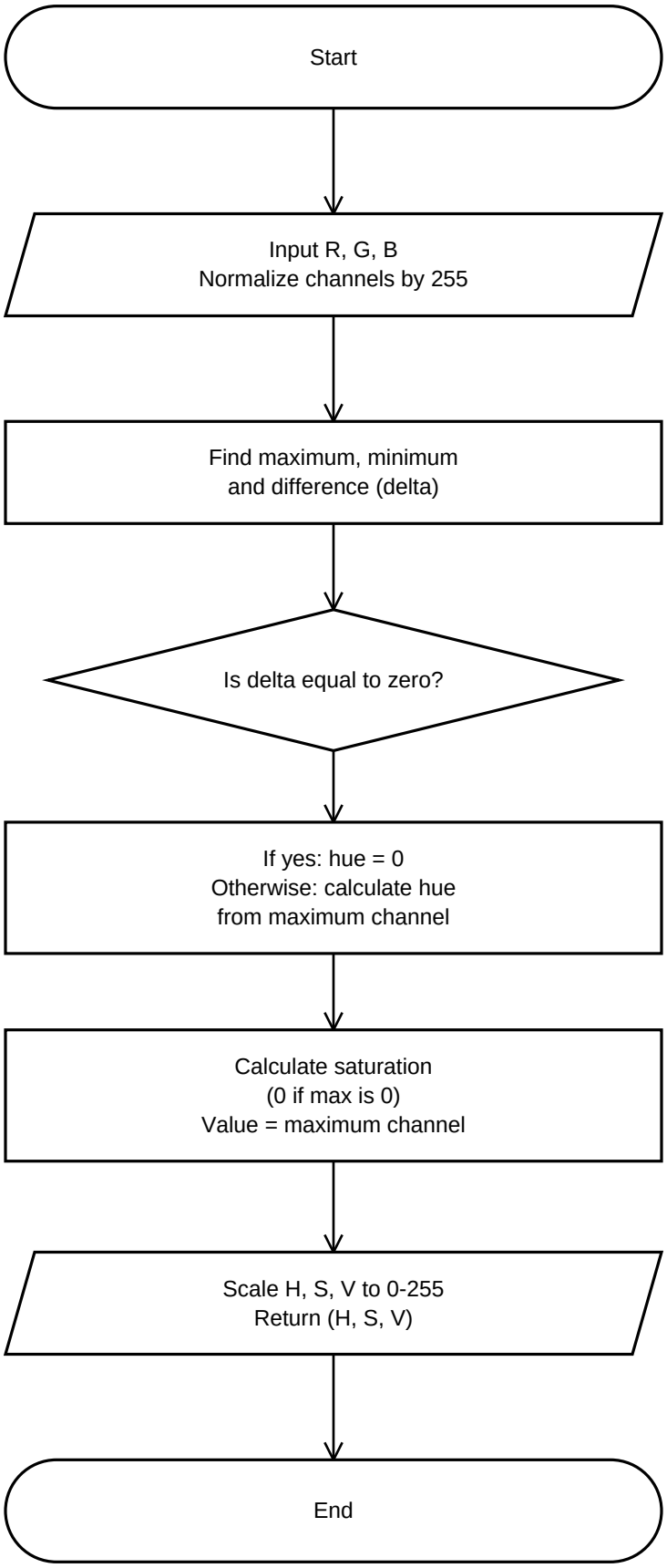
2. resize_image



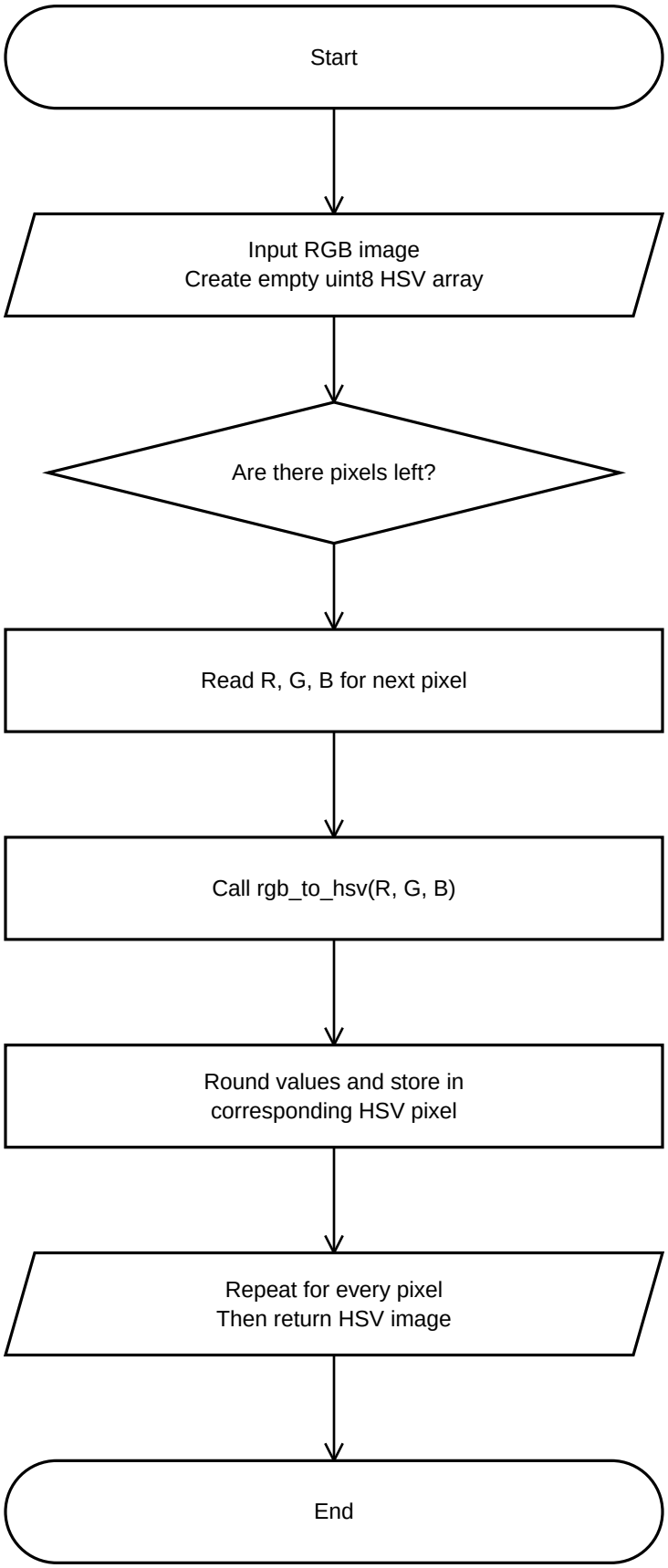
3. pad_image


Start
Read resized image dimensions
and target size
Calculate horizontal and
vertical padding amounts
Split padding between
opposite sides
Use NumPy edge padding
to extend border pixels
Return padded image array
End

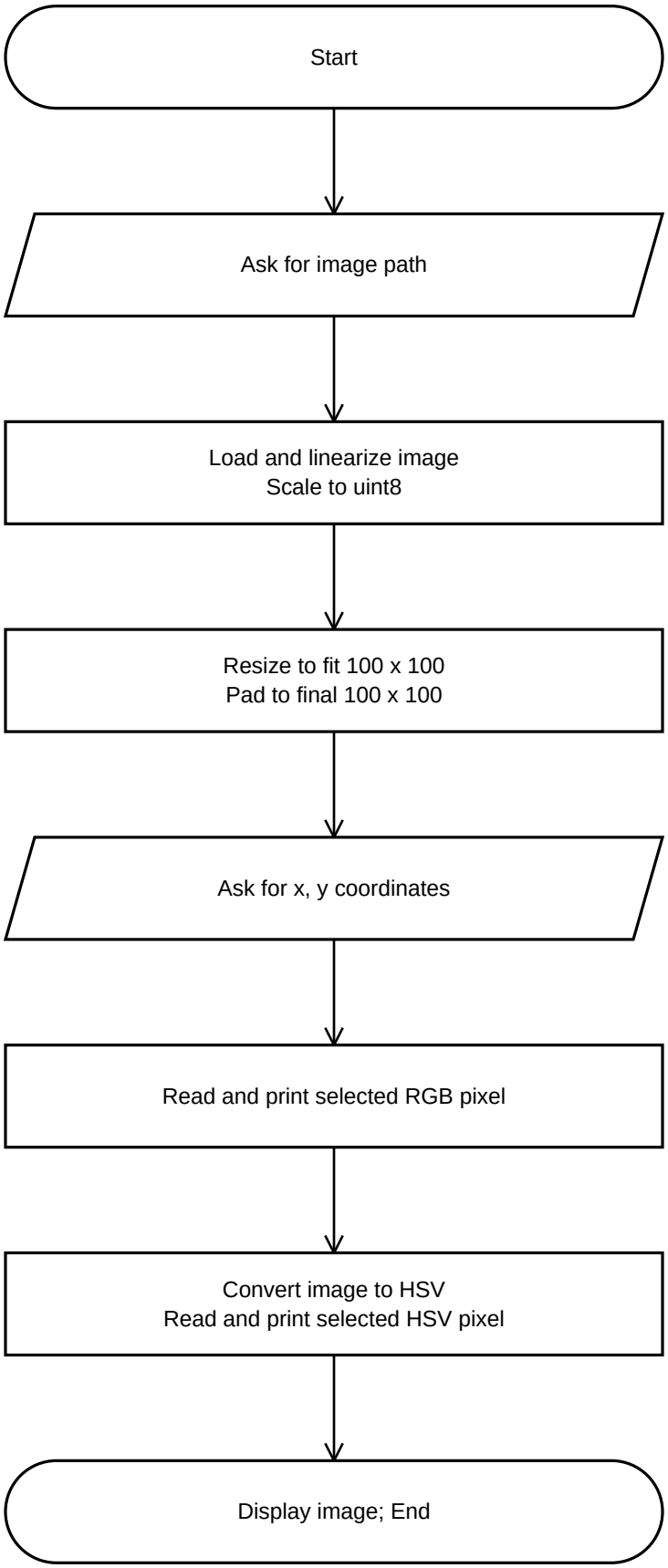
4. rgb_to_hsv


Start
Input R, G, B
Normalize channels by 255
Find maximum, minimum
and difference (delta)
Is delta equal to zero?
If yes: hue = 0
Otherwise: calculate hue
from maximum channel
Calculate saturation
(0 if max is 0)
Value = maximum channel
Scale H, S, V to 0-255
Return (H, S, V)
End

5. convert_to_hsv



6. main


Start
Ask for image path
Load and linearize image
Scale to uint8
Resize to fit 100 x 100
Pad to final 100 x 100
Ask for x, y coordinates
Read and print selected RGB pixel
Convert image to HSV
Read and print selected HSV pixel
Display image; End